Mustafa Siddiqui

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Education

University of South Florida (USF) — Aug 2026 – Present
Ph.D. in Computer Science — *Tampa, FL, USA*

Lahore University of Management Sciences (LUMS) — Aug 2019 – May 2023
B.S. in Electrical Engineering (Graduated with Distinction) — *Lahore, Pakistan*

Research Interests

Knowledge Editing in Large Language Models, Mechanistic Interpretability, Neural Knowledge Representation, Test-Time Adaptation, Generalization Theory, Foundations of Deep Learning

Publications

Multi-Algorithm Deep Unfolding With Application to Foreground Detection. *IEEE Signal Processing Letters*, 2025. IEEE

Scalable Ensemble Federated Learning with Enhanced Open-Set Recognition. *Transactions on Machine Learning Research*, June 2026. tmlr

Experience

Research

Graduate Research Assistant, USF — *Aug 2026 – Present | Tampa, FL*

- **Knowledge Editing:** Developed a Description Logic-based benchmark to investigate how large language models represent, retrieve, and modify factual knowledge, distinguishing assertions (ABox), concepts (TBox), and relations (RBox). The benchmark examines the limitations of parameter editing, particularly for relational knowledge, and whether existing methods achieve genuine belief revision or merely alter prompt-specific responses.

Research Assistant, LUMS — *Jul 2023 – Jul 2026 | Lahore, Pakistan*

- **Open-Set Recognition:** Conducted extensive empirical and theoretical analyses of open-set recognition methods, identifying key performance factors, fundamental limitations of existing objectives, and unifying the major perspectives and motivations in the literature within a common framework.
- **Federated Learning:** Developed FedSOV, a scalable ensemble-based federated learning framework leveraging open-set recognition and model pruning to address extreme data heterogeneity. Conducted theoretical analysis comparing parameter-space and function-space aggregation.
- **Domain Generalization:** Investigated the relationship between optimization dynamics, polysemantic neural representations, and domain-specific feature learning. Developed optimization strategies to discourage reliance on unstable features and improve generalization across domains.
- **Deep Unfolding:** Proposed a multi-algorithm deep unfolding framework that enhances neural network expressivity through proximal diversity, validated on foreground detection.
- **Combinatorial Optimization:** Developed heuristics for the Capacitated Vehicle Routing Problem (CVRP) and integrated them into a dashboard for the Lahore Waste Management Company.

Teaching

Graduate Teaching Assistant, USF — *Aug 2026 – Present | Tampa, FL*

- Teaching Assistant for Network Security and Firewalls, supporting Capture the flag / Network protection laboratory exercises and assignment evaluation.

Teaching Assistant, LUMS — *Aug 2022 – May 2023 | Lahore, Pakistan*

- Assisted in teaching Electromagnetic Fields and Waves and Feedback Control Systems through assignment design, grading, and student support.

AI Engineering

Freelance, Independent — *Jan 2025 – Jul 2026 | Remote*

- Developed AI applications including real-time voice agents, multi-agent banking systems, and graph-based retrieval augmented generation pipelines.